Slow Dynamics and High Variability in Clustered Networks

Reproduction of Litwin-Kumar and Doiron (2012)

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Abstract

This report reproduces the key findings of Litwin-Kumar and Doiron (2012) on slow dynamics in clustered cortical networks. We implement a network of leaky-integrate-and-fire neurons with increased connection probabilities based on neurons cluster assignment. We examine the effects of clustering on the network's firing patterns through measures like Fano factor, correlation coefficients, and firing rate distributions.

1. Introduction

Cortical networks exhibit slow temporal dynamics and high trial-to-trial variability that are difficult to explain with classical balanced network models.

1.1. Background

The balance between excitation and inhibition is a fundamental organizing principle of cortical circuits.

1.2. The Clustering Hypothesis

Clustered connectivity within excitatory populations can generate slow dynamics through metastable attractor states.

1.3. Objectives

We aim to reproduce the core findings of the original study and explore extensions to the model.

2. Model

We implement a recurrent spiking network with clustered excitatory connectivity using Brian2.

2.1. Network Architecture

The network consists of excitatory and inhibitory leaky integrate-and-fire neurons with structured connectivity.

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2.2. Neuron Model

Membrane potential dynamics follow a leaky integrate-and-fire model with exponential synaptic filtering.

2.3. Connectivity

Connection probability between excitatory neurons depends on cluster membership, with enhanced within-cluster connectivity.

2.4. Parameters

Model parameters are chosen to match those of the original study.

3. Results

We present our reproduction of the key findings from the original study.

3.1. Homogeneous Network Baseline

Networks without clustering exhibit stationary activity with low variability.

3.2. Clustered Network Dynamics

Introducing clustering produces slow fluctuations through transient cluster activations.

3.3. Quantitative Measures

We quantify temporal structure using Fano factor and autocorrelation analysis.

4. Extensions

Beyond reproduction, we explore extensions to the clustered network model.

4.1. Extension 1

Description of the first extension and its motivation.

4.2. Extension 2

Description of the second extension and its motivation.

5. Discussion

Our reproduction confirms that clustered connectivity generates slow dynamics through metastable attractor states.

5.1. Interpretation

The clustering mechanism provides a parsimonious explanation for slow cortical fluctuations.

5.2. Limitations

The model uses simplified neurons and imposed rather than emergent clustering.

5.3. Future Directions

Promising extensions include plasticity mechanisms and functional implications.

6. Conclusion

We have successfully reproduced the key results demonstrating that clustered connectivity generates slow dynamics in spiking networks.

References